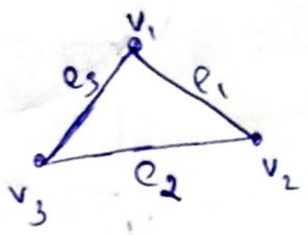
Q) Show that the complete graph K_5 is not planar. [4 marks]

A) The edge connecting V_2 & V_5 can be drawn outside and edge connecting V_3 and V_5 but the edge connecting V_1 and V_3 cannot be drawn outside because from outside the edges are crossed. Hence the edges ~~crossed~~. ~~Hence it~~ cross each other in a complete K_5 graph.

$\therefore$ It is not planar.

Walk:

A walk in a graph is defined as a finite sequence of vertices and edges arranged alternatively.



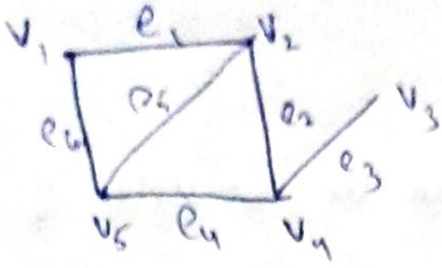
$v_1 e_1 v_2 e_2 v_3 e_3 v_1$

~~Length~~: length of walk = No. of edges in a graph.

length of $W = 3$.

Trail:

A walk with distinct edges is called trail. i.e., A walk with non repeated edges. ~~is called~~



Trail - $v_1 e_1 v_2 e_2 v_4 e_4 v_5 e_6 v_1$

Path:

Path is a walk with distinct vertices ie., from the above figure

- $v_1 e_1 v_2 e_5 v_5 e_6 v_1$ is a path.
- $v_1 e_1 v_2 e_2 v_4 e_3 v_3$ is a path.

Cycle:

A closed path is a cycle, no edges is repeated and the starting and ending vertex are same.

Circuit.

* A closed trail is a circuit. Edges are not repeated. Vertices may repeat.
* If all the vertices are distinct from the initial vertex to terminal vertex then the walk is called circuit.
* Every cycle is a circuit, But not every circuit is a cycle.

Isomorphism

Two graphs G_1 and G_2 are said to be isomorphic if:

i) they have same number of vertices.

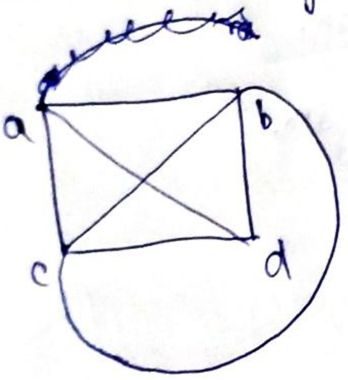
ii) same number of edges.

iii) same degree for corresponding vertices.

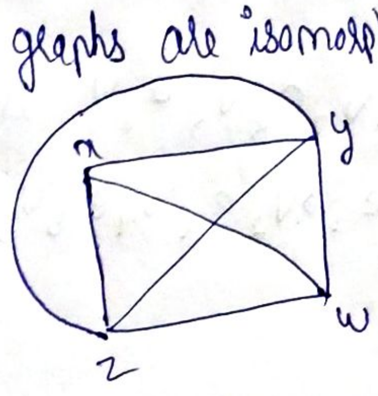
iv) same number of loops.

We write, ~~G_1 is~~ $G_1 \cong G_2$.

-> Check whether the following graphs are isomorphic or not.



G_1



G_2

$f(a) = w$

$f(b) = z$

$f(c) = y$

$f(d) = x$

edge correspondence

$\{a,b\} = \{z,w\}$

$\{a,c\} = \{y,w\}$

$\{c,d\} = \{x,y\}$

$\{b,d\} = \{x,z\}$

$\{c,b\} = \{z,y\}$

$\{a,d\} = \{x,w\}$

$d(a) = 3 = d(u)$

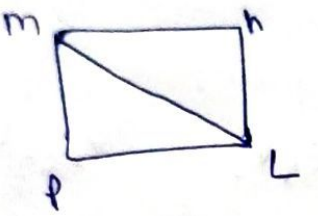
$d(b) = 4 = d(z)$

$d(c) = 4 = d(y)$

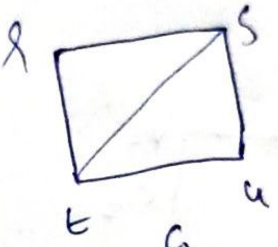
$d(d) = 3 = d(x)$

Clearly it is isomorphic

→ check whether the graphs G_1 and G_2 are isomorphic



G_1



G_2

$f(m)$

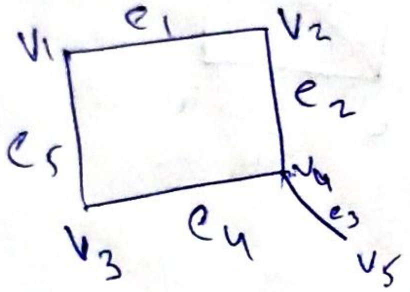
H.W

Euler graph.

If a walk contains all the edges of a graph exactly once then the walk is called Euler trail.

If there is a Euler circuit in the graph we call it Euler line

example :-



Euler circuit

$V_4 e_4 V_3$ ---- $V_2 e_2 V_4 e_3 V_5$

HAMILTONIAN GRAPH

Hamiltonian graph is a simple circuit that ~~passes~~ passes through every vertex exactly once.